$$\bar{x} = \frac{7+9+12+15+5+4+11}{7}$$

$$= \frac{63}{7} = 9$$

Hence the mean of the number of books is 9

(ii) Short-cut Method to find A.M.

Under this method an assumed mean or an arbitrary value (denoted by A) is used as the basis of calculation of deviations (d_i) from individual values. That is if $d_i = x_i - A$ then

$$\bar{x} = A + \frac{\sum_{i=1}^{n} d_i}{n}$$

Example 5.2

A student's marks in 5 subjects are 75, 68, 80, 92, 56. Find the average of his marks.

Solution:

Let us take the assumed mean, A = 68

x_i	$d_i = x_i - 68$
75	7
68	0
80	12
56	-12
92	24
Total	**31**

$$\bar{x} = A + \frac{\sum_{i=1}^{n} d_i}{n}$$

$$= 68 + \frac{31}{5}$$

$$= 68+6.2 = 74.2$$

The arithmetic mean of average marks is 74.2

(b) To find A.M. for Discrete Grouped data

If $x_1, x_2, \ldots, x_n$ are discrete values with the corresponding frequencies $f_1, f_2, \ldots, f_n$. Then the mean for discrete grouped data is defined as (direct method)

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{N}$$

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